



Scope of machine learning in materials research—A review

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ABSTRACT

This comprehensive review investigates the multifaceted applications of machine learning in materials research across six key dimensions, redefining the field's boundaries. It explains various knowledge acquisition mechanisms starting with supervised, unsupervised, reinforcement, and deep learning techniques. These techniques are transformative tools for transforming unactionable data into insightful actions. Moving on to the materials synthesis, the review emphasizes the profound influence of machine learning, as demonstrated by predictive models that speed up material selection, structure-property relationships that reveal crucial connections, and data-driven discovery that fosters innovation. Machine learning reshapes our comprehension and manipulation of materials by accelerating discovery and enabling tailored design through property prediction models and structure-property relationships. Machine learning extends its influence to image processing, improving object detection, classification, and segmentation precision and enabling methods like image generation, revolutionizing the potential of image processing in materials research. The most recent developments show how machine learning can have a transformative impact at the atomic level by enabling precise property prediction and intricate data extraction, representing significant advancements in material understanding and innovation. The review highlights how machine learning has the potential to revolutionize materials research by accelerating discovery, improving performance, and stimulating innovation. It does so while acknowledging obstacles like poor data quality and complicated algorithms. Machine learning offers a wide range of exciting prospects for scientific investigation and technological advancement, positioning it as a powerful force for influencing the future of materials research.

Introduction

Machine learning has become a revolutionary tool within materials science research, allowing for quick and precise analysis of complicated material properties, predicting novel materials with desired properties, and optimising fabrication procedures. With previously unseen insights into the molecular and atomic levels of material behavior, its scope also includes improving material discovery, design, and performance

prediction [1]. Machine learning techniques have enormous potential to revolutionize how we comprehend, work with, and innovate in materials science as they develop [2].

Machine learning includes various techniques that allow computers to learn and make wise decisions [3]. Models are learned from data with labels through learning under supervision, which enables them to make precise predictions [4]. Unsupervised learning finds structures and patterns in unlabeled data, bringing to light hidden information [5].

Abbreviation: ML, machine learning; RL, reinforcement learning; BPSM, bayesian propensity score matching; NWP, numerical weather prediction; SVR, support vector regression; SVM, support vector machine; SRA, structural reliability analysis; HEAs, high-entropy alloys; DFT, density functional theory; BMGs, bulk metallic glasses; HTE, high-throughput experimentation; HTS, high-throughput screening; MVMOO, mixed variable multi-objective optimization; VAEs, variational autoencoders; GANs, generative adversarial networks; ANFIS, adaptive neuro-fuzzy inference system; MSE, mean squared error; GA, genetic algorithms; FNO, frontier orbital nature; HOMO-LUMO, highest occupied molecular orbital–lowest occupied molecular orbital; CNOHF, carbon nitrogen oxygen hydrogen fluorine; LOOCV, leave-one-out cross-validation; AI, artificial intelligence.

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Through trial and error, reinforcement learning teaches models to choose actions to maximize rewards [6]. Deep understanding, a subset, uses neural networks to process complex data hierarchies for complex tasks like image and speech recognition [7]. Transfer learning occurs when knowledge is transferred from one job to another, increasing productivity [8]. Probabilities are included in Bayesian methods for modelling uncertainty [9]. For improved accuracy, ensemble methods combine multiple models [10]. The best decision boundaries are produced by Support Vector Machines [11]. Together, these methods drive innovation across industries, advancing artificial intelligence capabilities.

Machine learning has sparked a revolution in the synthesis of materials by enabling a diverse array of uses. By using models to predict material behaviors, property prediction speeds up the selection of materials for particular functions [12]. Intricate connections between material structures and desired properties are revealed by structure-property relationships, enabling targeted design [13]. Efficiency is increased by high-throughput screening, which expedites the evaluation of large material libraries [14]. For best results, process optimization optimizes fabrication parameters [15]. Innovation is accelerated by data-driven material discovery, which finds promising candidates in sizable datasets [16]. In material synthesis, generative models produce novel materials with specialized properties that inspire creativity and exploration [17]. Together, these applications redefine the field of materials science, promoting rapid advancement and discovery.

Material predictions across domains are revolutionized by machine learning. Property prediction models accurately predict the behaviors of materials, and materials discovery and design quickly produce novel compounds tailored for particular tasks [18]. Critical connections between material structures and desired traits are revealed by structure-property relationships [19]. Manufacturing conditions are adjusted through process optimization, increasing productivity [20]. Extensive material libraries are quickly evaluated by high-throughput screening [21]. Thanks to material characterisation and data analysis, materials science is advanced through accelerated innovation [22].

Machine Learning has transformed image processing, propelling a diverse array of uses. While object identification and localization precisely locate and identify objects within images, image classification accurately labels the images [23]. Understanding is improved by the precise boundaries that are drawn by image segmentation and object recognition. Techniques for strengthening and restoring images improve image quality [24]. Creating lifelike visuals by image generation and synthesis inspires creativity [25]. Understanding and analyzing images delves into intricate patterns for insights [26]. These developments collectively rethink how we process, manipulate, and derive meaning from visual information, revolutionizing the entertainment and healthcare industries.

Machine learning innovations in recent years have opened up new horizons for materials research. Microscopic property prediction uses models to predict materials' atomic-scale behaviors, enabling previously unheard-of precision [27]. Complex algorithms are used in material characterization and imaging to extract fine details from extensive data [28]. Data-driven insights are used in material property analysis to unearth hidden correlations and trends [29]. Material discovery and design are sped up through predictive models, producing novel compounds with customized properties quickly [30]. These innovations usher in a transformative era where machine learning reshapes how we perceive and manipulate materials and fuel innovation across diverse fields.

While machine learning in materials research has enormous potential, there are also difficulties. Obtaining high-quality labelled data and integrating domain knowledge into models remain challenges [31]. For precise predictions, complex material behaviors and intricate relationships necessitate sophisticated algorithms [32]. Black-box models' interpretability and generalization from sparse data raise questions

[33]. It is crucial to consider data usage and bias reduction ethical considerations [34]. Despite these obstacles, machine learning promises quick material discovery, optimization, and innovation [35]. It holds the key to unlocking previously unattainable insights and accelerating materials research, ushering in a new era of scientific exploration and technological advancement by overcoming challenges through interdisciplinary collaboration and algorithmic improvements.

The objective of this review article is to offer a comprehensive exploration of the utilization of machine learning in materials research. The first section discusses various machine learning methodologies. The utilization of machine learning in the synthesis of materials is discussed. The substantial contribution that machine learning makes to the prediction of material properties is covered in the following section. The following sections examine recent developments in materials research and the utilization of machine learning to image processing. The paper acknowledges the challenges and prospects in its concluding section. It highlights the need for ongoing research and development to ensure the effective integration of machine learning while realising its exciting potential to influence the direction of materials research.

Machine learning methods

By utilizing massive datasets and machine learning (ML), which enable computers to learn as they progress, programming has removed many dangers and dead ends [36]. These algorithms can recognize patterns, extrapolate from examples, and change over time [37]. In the literature, various machine-learning approaches are applied to risk assessment [38]. A subfield of computer science called "machine learning" seeks to extract patterns from data to improve performance across various activities [39]. Fig. 1 shows different methods of machine learning.

Supervised learning

Specific objectives are frequently considered during the design process for materials and machines to respond as desired to applied forces or limitations [40]. Supervised learning stands as the predominant form of machine learning. It operates in two steps. The algorithm identifies patterns within the training dataset, consisting of sample sets and labels. Then it transforms these patterns into a mathematical expression known as a model during the training step. This model is utilized to make predictions about samples it hasn't encountered during the inference phase [41]. New technologies, such as Google TPUs and NVIDIA CUDA, have emerged to provide faster data processing as the quantity and size of datasets accessible for research and commercial purposes continue expanding exponentially. [42,43]. Over the past few decades, computation speeds have been rising quickly in 1 [arXiv:2006-12-25](#) Version 1 (quant-ph) Moores Law, which states that the number of transistors in a compact integrated circuit (IC) doubles roughly every two years, will come into effect on June 22, 2020 [44]. Various techniques provide a function that assigns inputs to the desired corresponding outputs during learning with supervision. The classification problem is a typical example of a learning task with supervision. In this scenario, the learner must approximate the behavior of a function that assigns a vector to one of several classes. This is achieved by analyzing numerous input-output samples of the function [45]. The learning process of a basic machine learning model involves two stages: training and testing. In the training stage, the model is constructed using examples from the training data as input, allowing the learning algorithm or learner to acquire features [46]. Supervised learning is used when labeled data is available for training. It's applied in classification (e.g., spam detection, image classification) and regression (e.g., predicting house prices) tasks [47].

In this learning, models are trained on labeled data, learning the mapping between input features and their corresponding outputs. Algorithms like Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVMs), and Neural

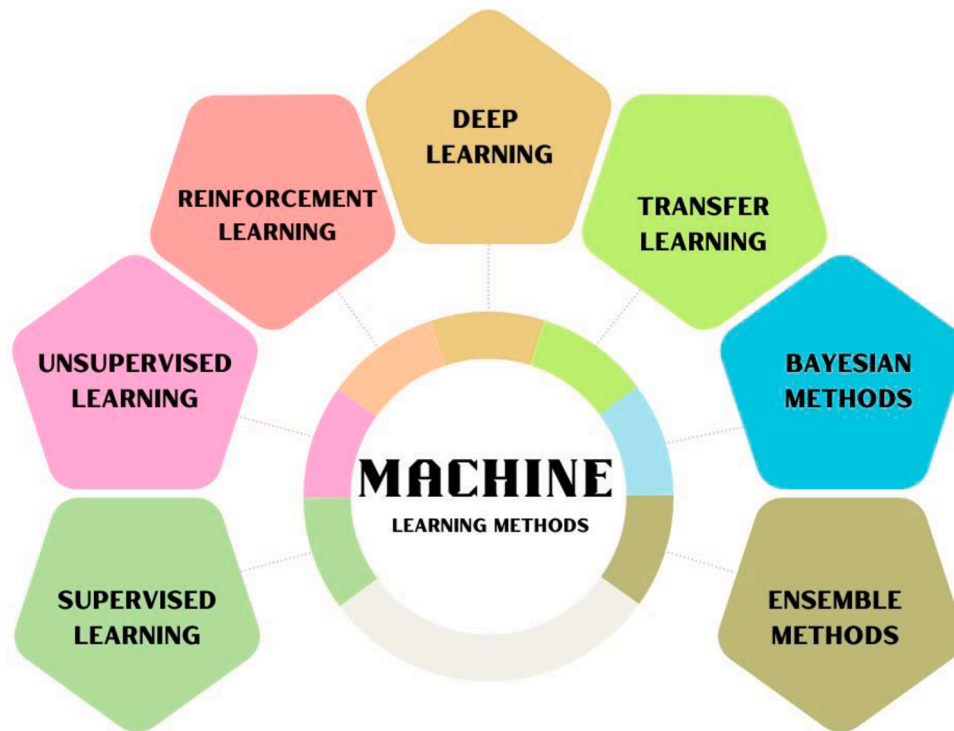


Fig. 1. Methods of machine learning.

Networks (in a supervised context) fall under this category [40]. Fig. 2 illustrates the flowchart of the supervised learning process.

The Support Vector Machine (SVM) is a renowned learning algorithm for solving classification and regression problems which is also a kind of Supervised Learning. The complexity and effectiveness of prediction models are significantly influenced by SVM parameters like kernel and penalty parameters [48]. A potent machine learning method that applies the principle of minimizing structural risk is called (SVM). In recent years, there has been a significant surge in the application of Support Vector Machines (SVM) for conducting Structural Reliability Analysis (SRA) [49]. The Support Vector Machine (SVM) has the potential to enhance accuracy and generalization when estimating implicit functions, owing to its foundation in the structural risk reduction concept and its adeptness at handling small sample learning [50,51]. SVM has shown superior performance compared to SVR due to its ability to mitigate overfitting in regression and its improved response approximation capabilities [52,50]. SVMs work well in scenarios where there's

a clear margin of separation between classes. They are effective for classification tasks, especially in high-dimensional spaces, such as text categorization, image classification, and bioinformatics [53].

SVMs are supervised learning models used for classification and regression analysis. They find a hyperplane that best divides a dataset into different classes. SVMs use a kernel trick to transform data into higher dimensions for better separation when the data is not linearly separable [54].

Y. Liu, Wang, et al. developed Auto-MatRegressor, an automated method using meta-learning for predicting material properties. It streamlines accurate ML model building in materials science by automatically choosing algorithms and optimizing parameters. It utilizes 27 dataset meta-features and performance metrics of 18 common algorithms, incorporating domain knowledge via a materials categories tree. Through experiments on 60 datasets, Auto-MatRegressor efficiently selects algorithms, outperforming traditional methods while being adaptable to various materials discovery and design tasks [55].

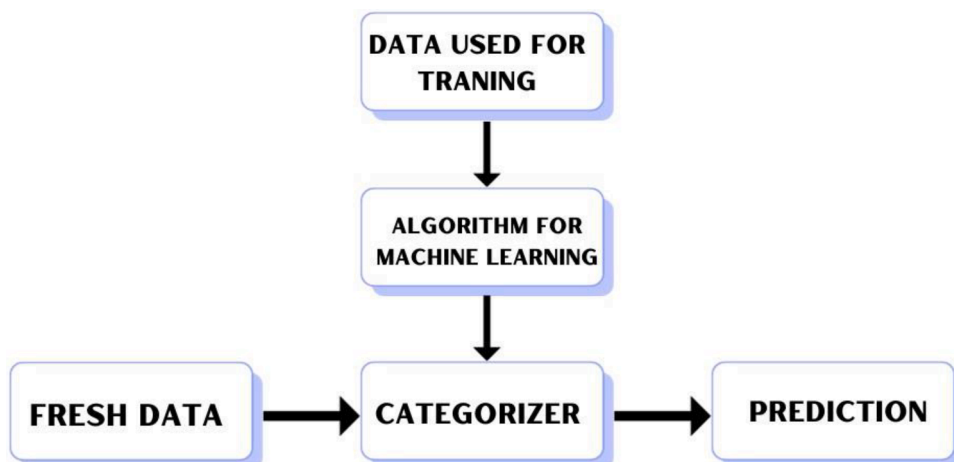


Fig. 2. The process of learning with supervision.

Unsupervised learning

In material research, solid-state physics, biophysics, and biochemistry, examining the extensive quantities of data produced by atomistic and molecular simulations is becoming increasingly important [56]. The performance gap between supervised pretraining and unsupervised visual representation learning, also known as self-supervised learning, in computer vision is quickly shrinking. Self-supervised learning tries to acquire features without manual annotations [57]. Unsupervised learning is useful when dealing with unlabeled data to find hidden patterns or structures within the data. Clustering (like K-means), dimensionality reduction (PCA, t-SNE), and generative modeling (GANs) are examples applicable to various domains such as customer segmentation, anomaly detection, and data compression [58].

Algorithms in this category aim to find hidden patterns or intrinsic structures in data without explicit supervision. Examples include Clustering algorithms like K-Means, Dimensionality Reduction methods like Principal Component Analysis (PCA), and Generative Adversarial Networks (GANs) for generating synthetic data [59]. Fig. 3 devotes the uses of unsupervised learning in different fields.

Reinforcement learning

For learning-based control, reinforcement learning offers a mathematical formalism. We can automatically develop near-optimal behavioral abilities, represented by rules, for improving user-specified reward functions using reinforcement learning [60]. In recent years, deep reinforcement learning (RL) methods have driven significant breakthroughs in artificial intelligence, surpassing human performance in games such as Atari, Go, and no-limit poker [61]. Central to reinforcement learning (RL) is acquiring a behavioral policy that maps states or situations to actions to maximize the cumulative long-term reward [62]. Deep neural networks and RL were not combined until 2015 when groundbreaking research showed how to make deep RL effective in challenging domains like Atari video games [63]. Since then, deep RL has made quick strides toward scaling and improvement, enabling its use in complex tasks like Go and Capture the Flag [64,65]. The latter advancements frequently entailed combining deep reinforcement learning (RL) with architectural and algorithmic complements, including tree search or using slots for episodic memory [66,67]. Reinforcement learning is applied in scenarios where an agent learns to make sequences of decisions by interacting with an environment. It's useful in robotics, game playing (like AlphaGo), autonomous vehicle control, and optimizing processes, where learning through trial and error is necessary [68].

This involves training agents to make sequences of decisions in an environment to maximize cumulative reward. Algorithms like Q-

Learning, Deep Q Networks (DQN), Policy Gradient Methods, and Actor-Critic methods are commonly used in reinforcement learning [69].

Deep learning

Deep learning consists of interconnected classifiers built on linear regression and a few activation functions [70]. Shallow learning and deep learning are the two distinct stages of machine learning development. Shallow learning will be restricted to no more than a pair of nonlinear feature conversion layers, as opposed to deep learning's limit of four. The most prevalent shallow structures are Gaussian Mixture Models, Logistic Regression, and Support Vector Machines [71–80]. Deep learning's theoretical understanding of the connection between performance, model complexity, and computational needs is still lacking. However, there are compelling arguments for thinking that deep learning depends on processing capacity. This results from the over-parameterization problem and how it worsens as more training data are used to boost performance [81]. Deep learning is well-suited for problems with large amounts of data, such as image recognition, natural language processing, speech recognition, and other complex pattern recognition tasks. It excels in learning hierarchical representations from raw data [82]. This encompasses a broad range of algorithms that use artificial neural networks with multiple layers to extract hierarchical representations from data. Some popular deep learning algorithms include Convolutional Neural Networks (CNNs) for image-related tasks, Recurrent Neural Networks (RNNs) for sequential data, and Transformer models like GPT (Generative Pre-trained Transformer) for natural language processing [83].

Transfer learning

While traditional machine learning technology has achieved significant success and found practical applications, it still possesses inherent limitations when applied to specific real-world situations. Transfer learning, a compelling approach in machine learning, offers a solution for the mentioned challenge by leveraging knowledge from one domain to another. The concept of transfer learning is rooted in educational psychology, suggesting this field initially influenced it [84]. There are various transfer learning categorization standards. Transductive, inductive, and unsupervised transfer learning are three categories that categorize transfer learning issues [85]. Transfer learning is beneficial when you have a small dataset in the target domain but a large dataset in a related source domain. For instance, in image recognition, you can use a pre-trained model on a large dataset (like ImageNet) and fine-tune it on a smaller dataset specific to your problem [86]. This technique involves leveraging knowledge from one domain or task to aid learning in another. Algorithms like Fine-tuning pre-trained models (e.g., in NLP -

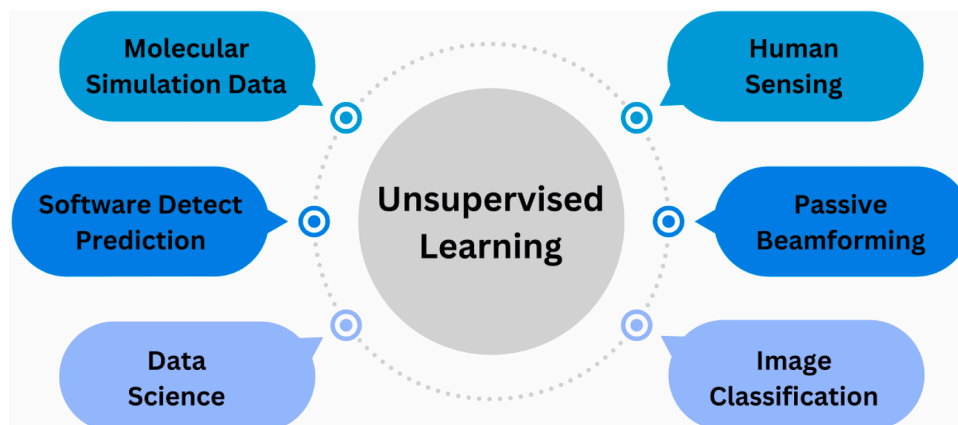


Fig. 3. Different fields of unsupervised learning.

BERT, GPT, etc.) and Domain adaptation (adapting a model trained on a source domain to a target domain) are commonly used in transfer learning [87].

Bayesian methods

The arrangement of parameters in the population is generated using a simulation-based method through the Bayesian approach in the current study. The ability of Bayesian Propensity Score Matching (BPSM) to account for uncertainty reduces the arbitrary nature of decisions regarding subclassification in the matching process [88]. In the past, Bayesian inference—which offers precise, complete prediction distributions and precisely calibrated uncertainty—was the gold standard for learning using neural networks. Because of the vast complexity of the parameter space, it isn't easy to scale Bayesian inference methods to deep neural networks [89]. Bayesian methods are used to model uncertainty and update beliefs based on evidence. They're suitable for problems involving decision-making under uncertainty, probabilistic reasoning, and in cases where prior knowledge or assumptions need to be incorporated into the model, like in medical diagnosis, natural language processing, and financial modeling [90].

These methods involve using probability to model uncertainty. Naive Bayes is a simple and popular algorithm, especially for text classification. Bayesian Networks and Bayesian Linear Regression are other examples that leverage Bayes' theorem to update beliefs based on new evidence [91].

Ensemble methods

To address the shortcomings of single prediction models, ensemble models have gained popularity recently. Ensemble modelling uses data variability to make up for component model flaws like bias and produce a more reliable solution and resistance to Numerical Weather Prediction (NWP) model mistakes [92]. Numerous researchers have introduced various methods for forming an ensemble. The foundation of one solution (sol1) was obtaining several training data sets. The authors discussed the utilization of bagging and boosting techniques to create such data, and they highlighted that decision tree-based methods are models that use this type of data. Another approach (sol2) included employing various ensemble members as predictive model members [93]. Machine learning models are becoming routinely used prediction tools, both as solo solutions and as ensemble members. Two frequently used prediction methods are neural networks and decision trees, along with different variants of Support Vector Regression (SVR) [94–97]. Ensemble methods combine multiple models to improve performance and generalizability. They're effective in situations where you want to reduce overfitting and increase accuracy, applicable in various domains like finance (predicting stock prices), healthcare (diagnosis), and recommendation systems [98].

This method combine multiple models to improve predictive performance. Examples include Bagging (Bootstrap Aggregating) with Random Forests, Boosting with algorithms like AdaBoost and Gradient Boosting, and Stacking where multiple models are combined using another meta-model [99].

Machine learning in materials synthesis

Machine learning (ML) for materials is rapidly transforming many fields to design innovative materials with higher performance [100] rapidly. Materials have a significant impact on how high technology is developed. Impressive advancements in creating new materials have been made due to the need for unique materials with optimum qualities. Big data may be created and accessed more quickly in the short term using experimental and computational methods. In the meantime, following the traditional eras of scientific advancements driven by experiments, theory, and computational simulations, data-driven research

has emerged as the fourth scientific paradigm, primarily due to the increasing scale of available data [101–103]. Machine learning (ML), a field combining statistical learning and computer science, has gained prominence recently as a rapidly evolving subfield of data-driven science [104,105]. ML gives us a new strategy for finding new materials distinct from the conventional method of experimentation involving trial and error. To provide direction for material design, machine learning (ML) makes new predictions by comprehending the data's latent patterns [106]. The creation of numerous top-performing alloy materials, such as titanium alloy, nickel alloy, high-entropy alloys (HEAs), aluminium alloy, shape memory alloy, magnesium alloy, bulk metallic glass alloys (BMGs), nickel-based single crystal superalloys [107,108], and copper alloy, has heavily relied on machine learning (ML) over the past ten years [109–115]. For example, although bulk metallic glasses (BMGs) exhibit unique features like high mechanical strength, low magnetic hysteresis, and strong corrosion resistance, it remains crucial to establish dependable methods for correlating these desired traits with specific compositions [116,117].

The utilization of machine learning in the development of novel materials has expanded. It presents fresh chances to quicken and improve searching for novel materials with desired features. Fig. 4 shows examples of how artificial intelligence is used in material synthesis.

Property prediction

In machine learning, the term "material prediction" refers to using machine learning algorithms to forecast materials' characteristics, actions, or results. This method entails evaluating and learning from vast quantities of material data to create models that can generate precise predictions about the characteristics of novel and untested materials. The capability to detect molecules or compounds exhibiting one or more features within a predetermined range makes machine learning-based surrogate models especially well-suited for high-throughput screening endeavours. Discovering "optimal" combinations corresponds to finding chemistries on or near the underlying Pareto front. This front represents



Fig. 4. Artificial intelligence in material synthesis.

the best achievable trade-offs among conflicting reactions when a subset of characteristics displays opposing trends or inverse relationships [118]. For a wide range of material classes, including polymers, metals and alloys, 2D materials, ceramics and composites, organic-inorganic hybrids, and multi-component hetero-anionic compounds, machine learning algorithms have been used to find possible non-linear connections involving multiple variables [119–121]. Instances of predicting materials properties with success using machine learning (ML) include estimation of energetics [122–126], stability of phases and arrangement of cations/anions [127–133], defect energetics [134–137], bandgaps [138–142], temperatures at which melting and glass transition occur [143,144], mechanical and elastic properties [145–147], thermal conductivity [148], dielectric properties [149,150], tendency for crystallization [151], catalytic activity [152,153], and radiation damage resistance [154].

Predictive modelling techniques applied to Density Functional Theory (DFT) computations have expedited the discovery of new materials, significantly reducing the search space for upcoming DFT computations and experiments [155]. Since design codes frequently require these qualities, accurate prediction of the mechanical properties of concrete has been an issue. Researchers are pursuing accurate models for forecasting mechanical strength due to the introduction of novel concrete mixtures and applications [156]. Machine learning has a diverse array of uses to predict their property, as shown in Fig. 5.

Structure-property relationship

By utilizing self-organizing maps like Kohonen to show structure/property correlations between silver and platinum nanoparticles, data visualization helps comprehend materials datasets and successfully present results [157]. A suitable machine learning approach is utilized to present high-dimensional data within a lower-dimensional framework to make comparing structural features with attributes easier. One such method is the nonlinear dimension reduction method t-SNE which efficiently distinguishes between distinct and comparable items by modelling high-dimensional objects as two- or three-dimensional points [158]. Biological applications and displaying representations from artificial neural networks are only two examples of the many disciplines

in which it has been effectively used [159–161]. The Self-Organizing Map (SOM) or Kohonen network [162] is a dimensional reduction technique for materials datasets. It is an unsupervised artificial neural network [163] that preserves the topological connection of the input data while reducing high-dimensional spaces to lower-dimensional regions through mapping. SOMs have been used to present data with multiple dimensions as a single two-dimensional plot for further analysis and generating surface texture maps of nanoparticles as unique material identifiers [164]. To identify structure/property connections in metallic nanoparticles, the article contrasts t-SNE with SOMs. SOMs are proven to be more trustworthy and comprehensible for typical materials research datasets, avoiding distortion [157].

Machine learning algorithms can analyze large datasets of material structures and their accompanying attributes to establish correlations between them. The need for energy has increased in significance on a global scale with the growth of contemporary society. Therefore, there is a pressing need to investigate novel materials for sustainable energy systems. Extended experimental duration and significant expenses make traditional approaches challenging for materials research. Machine learning (ML) is currently emerging as a novel approach in research to transform the discovery of novel materials [165]. Many problems facing the 21st century today, such as sustainability and low-carbon energy, are primarily material-related. To ensure the sustainable growth of human society, materials with precise physical and chemical capabilities for adequate storage and conversion of energy are critically needed [166]. Artificial intelligence (AI) advancement creates fresh possibilities for scientific and engineering advances. Big data and artificial intelligence are "the fourth paradigm of science" [167]. As ML is a potent tool for uncovering the statistical rule concealed behind highly dimensional data, advancements in ML have recently significantly impacted several industries. With the recent expansion of materials databases, the progressive dissemination of ML toolkits like Tensor Flow, Pytorch, and sci-kit-learn, the creation of workflow toolkits like Atomate, and the advancement of algorithms, ML has also recently seen an increase in use in the area of materials science. ML approaches have effectively produced numerous improvements in materials for storing and converting energy, such as battery materials [168–172] and catalysts, by combining them with big data [173–183].

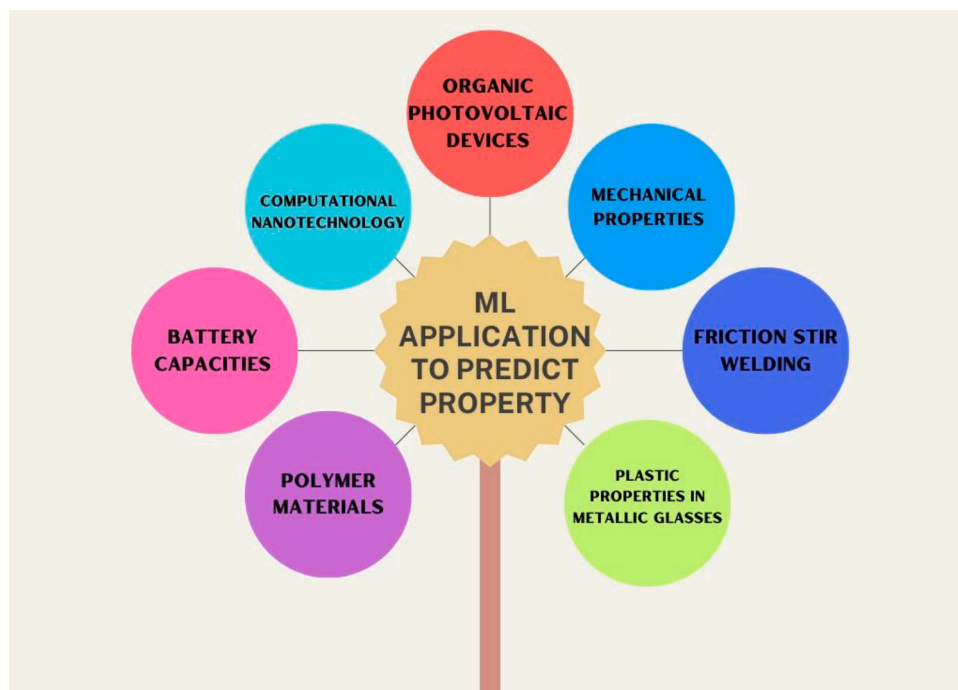


Fig. 5. Applications of machine learning for predicting properties.

High-throughput screening

In machine learning, high throughput screening (HTS) is a data-driven method for handling massive datasets, automating tests, refining models, and finding insightful information. It hastens research by quickly exploring complicated environments, offering data-driven conclusions, and making headway in many applications [184]. A low-cost method for incorporating transition-metal complexes and materials with open-shell configurations into current high-throughput screening techniques was demonstrated using machine learning (ML). To forecast border orbital energies and Frontier Orbital Nature (FNO) based diagnostics, various ML models were trained. These simulations demonstrated that the HOMO-LUMO gap is more ligand-sensitive, whereas the MR exhibits a higher metal sensitivity. These discoveries offer opportunities for individually modifying the two quantities [185, 186]. Density functional theory (DFT) is a key methodology for obtaining atomic electrical interactions. High-throughput screening is difficult due to the computationally costly nature of DFT calculations [187–189]. Machine learning (ML) techniques have recently been used to predict material properties [171,190–192] and molecular structures [193–195], significantly reducing the computational cost.

High-throughput studies, in which vast libraries of materials are produced and examined, can be guided by machine learning. According to recent literature, chemists and engineers have given machine learning (ML) and high-throughput experimentation (HTE) separate amounts of attention, which has resulted in the creation of potent reactivity models and platforms that can quickly carry out thousands of reactions [196]. Machine learning (ML) can enhance automated high-throughput chemistry testing in three domains: experimental design, route planning, and data processing. New research suggests that machine learning (ML) approaches hold significant potential in addressing the challenge of automated synthesis planning [197,198]. An expansive domain for exploring and enhancing catalysts results from the intricate interactions between various heterogeneous catalyst properties and operating variables that affect how well they work [199–201]. Trial & error testing has traditionally been used to overcome challenges like this, and it has been successful but time-consuming. The screening of many catalyst compositions for ammonia is a prime illustration. Synthesis carried out by Mittasch and colleagues using a preliminary use of high-throughput experimentation (HTE).

In general, HTE experiments are effective at efficiently screening a lot of catalysts. However, they can only "hit" a high-performing catalyst in the screened parameter space before they can make a discovery. To direct the tests, it is advantageous to combine HTE approaches with prediction algorithms [202,203]

Process optimization

In machine learning, process optimization entails using data-driven strategies to optimize processes, simplify workflows, and increase performance, efficiency, and effectiveness. Although complete kinetic schemes for both transport and kinetics in reacting particles are required in comprehensive models of biomass pyrolysis, they are computationally costly and not always feasible for producing sustainable fuels and chemicals [204]. To do this, it is necessary to create comprehensive prediction models that may be used to improve quick pyrolysis methods that turn lignin into valuable renewable fuels and chemicals [205]. As was previously mentioned, the kinetic information in the present comprehensive models is insufficient for this kind of optimization. It has successfully chosen the best model from various training models using dependable and efficient optimization strategies [206–208]. Much effort has gone into maximizing the parameters attained during training, including figuring out the best connection weights and minimizing the training error between projections and related goals. To enhance the precision of the independent Adaptive Neuro-Fuzzy Inference System (ANFIS), optimization approaches [209,210] are typically utilized for

ANFIS models to provide membership functions and related fuzzy rules. The optimum ML model configuration is discovered by reducing the Mean Squared Error (MSE) between target and predicted values using an integer-valued optimization technique based on Genetic Algorithms (GA) [211].

Machine learning and optimization techniques are integrated on the shop floor to enhance production processes. This is possible thanks to advancements in the manufacturing sector's digitalization process and the associated data availability. Additionally, a lack of resources encourages people to adopt new ideas more readily, like machine learning, which helps people conserve time, energy, and resources and prevent waste [212]. Process optimization in high-dimensional parameter space is required to provide a scalable manufacturing method for perovskite solar cells [213]. From discovering molecules and extending through process optimization to chemical manufacturing, machine learning is permeating all aspects of chemistry research and development [214–216]. This has caused the introduction of novel, challenging molecular targets with increased three-dimensionality, necessitating a more extensive and sophisticated reaction toolbox [217]. Constructing these more complex processes later is difficult and expensive, and doing so frequently necessitates optimizing both continuous and discrete variables. Before proceeding with the fine-tuning of continuous variables (such as reaction time, temperature, and stoichiometry), it's common to screen various combinations of discrete variables (such as catalyst, ligand, and solvent) using high-throughput screening (HTS) methods [218–221]. However, the time-consuming 'brute-force' approach to HTS (i.e., testing all potential combinations of discrete variables) is losing appeal due to increasingly complicated synthetic techniques. Additionally, since essential relationships among the combined variables are not considered, sequentially optimizing discrete and continuous variables in the workflow results in a limited understanding of the process [222]. Fig. 6 compares the traditional HTS method and newly developed self-optimization in the Mixed Variable Multi-objective Optimization (MVMOO) approach.

Data-driven material discovery

Materials discovery and design is a data-driven and systematic approach that uses cutting-edge computational tools, high-throughput screening and machine-learning methods to discover and enhance novel materials with specific desired characteristics [223–227]. This results in affordable and custom materials for various applications, propelling technological advancements. Datasets are required to aid in the detection of target traits, qualities, or unidentified materials when using machine learning to solve material issues. Researchers are turning to machine learning to supplement or partially replace the computationally and empirically costly DFT-based material discovery methodologies [227].

Machine learning in material discovery caters to both big and small datasets using distinct methodologies. With big datasets from materials genomics or high-throughput databases, methods like deep learning, ensemble techniques, and feature engineering thrive, harnessing the wealth of information to predict material properties efficiently. Dimensionality reduction techniques aid in handling numerous descriptors. In contrast, small datasets benefit from transfer learning's adaptation of pre-trained models, active learning, or Bayesian optimization to make data collection more efficient and improve model performance with limited data. Strategies like few-shot learning and meta-learning adapt to scarcity by leveraging knowledge from broader datasets, mitigating the challenges posed by limited data. Tailoring machine learning approaches to these scales optimizes material discovery processes, addressing complexities and scarcities inherent in big and small datasets [228–231].

Driven by the emergence of the big data era, data-centric research rapidly expands across diverse fields as it addresses the demand for faster and more systematic solutions to complex problems [175,232].

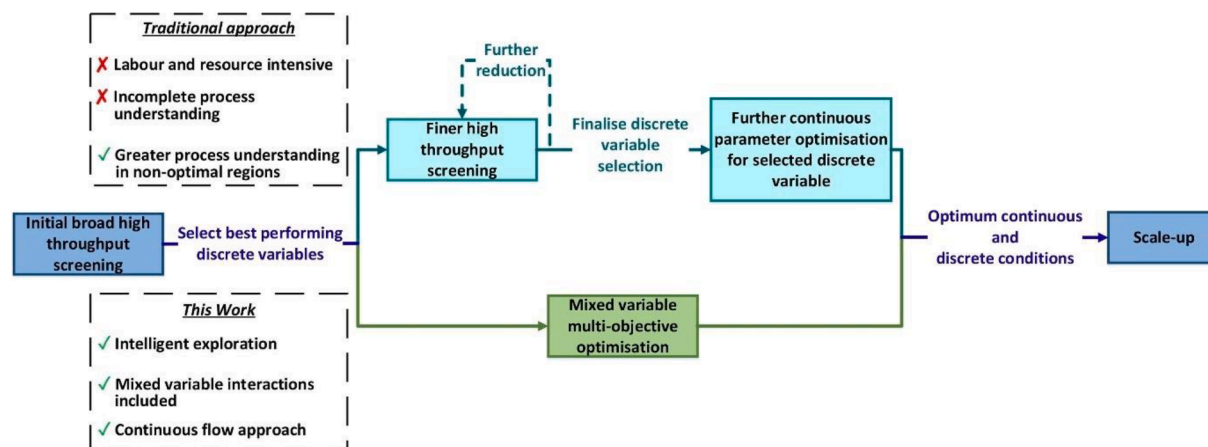


Fig. 6. Comparison of full reaction optimization using traditional HTS methodologies versus newly developed MVMOO self-optimization approach [222].

These techniques have delivered remarkable outcomes in catalysis, including discovering descriptors facilitating catalyst exploration or extracting patterns from spatially and temporally resolved spectroscopic data related to catalysts [233,234]. Machine learning shows the potential to achieve precision comparable to first-principles calculations while maintaining scalability similar to traditional computations [235]. Although there are many and varied applications of data science, catalysis, a discipline where optimization has traditionally relied heavily on trial and error and luck, offers particularly promising potential [236]. Large volumes of data are necessary for machine learning, and the lack of enough data, particularly experimental data, is the bottleneck of catalytic informatics. In this article, we go through three significant issues frequently faced in research focused on catalysis driven by data, specifically when supervised machine learning is utilized to forecast an experiment's results or the characteristics of a catalyst. These issues, particularly the absence of negative outcomes, are also significant in unsupervised machine learning and reinforcement learning, two other forms of machine learning [119].

In data-driven material discovery, several representative works, such as the Materials Project [237], AFLOW (Automatic FLOW for Materials Discovery) [238], and NOMAD (Novel Materials Discovery) [239], employ methodologies integrating computational materials science and machine learning. These projects harness high-throughput calculations, data aggregation, and predictive modeling to explore material properties efficiently. Within this context, Bayesian optimization emerges as a pivotal methodology facilitating the design process. It assists in optimizing experimental parameters, material compositions, or conditions by intelligently selecting the most informative experiments or simulations. Bayesian optimization operates by leveraging probabilistic models to guide the exploration of the vast parameter or design space, effectively accelerating the discovery of novel materials with desired properties. By employing Bayesian optimization techniques, these initiatives streamline the material discovery process, enabling the identification of promising materials for specific applications or functionalities [240].

Y. Liu, Yang, et al. employ data-driven machine learning (ML) for analyzing materials' structure-activity relationships and optimizing their performance, owing to ML's capacity to unveil hidden data patterns and provide accurate predictions. However, the challenge arises from a disparity between vast feature space and limited sample size, impacting ML model performance negatively. Efforts to tackle this include feature reduction, sample augmentation, and specialized ML techniques. Emphasizing a balance between sample size, features, and model parameters is crucial for managing data quantity. A proposed data quantity governance flow integrates materials domain knowledge, enhancing the process. By incorporating domain knowledge into governance schemes, this approach accelerates high-quality data acquisition, expediting materials design and discovery through ML

[241].

Generative models

Recent developments in deep generative models have facilitated the creation of highly realistic images. This development has prompted questions about the potential abuse of these models, though. The generated images may sometimes exhibit violent or graphic content or unintentionally expose sensitive data used during model training. A well-prepared dataset with adequate purification techniques might reduce the possibility of misusing generative models to address these problems [242]. For instance, as reported, creating DALLE-2 necessitated a meticulous selection of training data to exclude explicit content [243]. Machine unlearning could be used with data preparation and cleaning to help avoid issues when creating and deploying generative models. The objective of machine unlearning is to remove the target data from a pre-trained machine-learning model. This can be necessary to remove biased or damaging content and private information [244]. However, until now, most machine unlearning issues have been concentrated on supervised models [245–250].

Everybody's social media timelines appeared to be 50 years in the future in July 2019. Celebrities, including Drake and the Jonas Brothers, have been seen using the photo-filtering tool FaceApp to appear older. These filtered pictures—branded with the hashtag #AgeChallenge—started appearing on timelines and were frequently so unsettlingly accurate. Discussions regarding the ramifications of high-quality fake photos and data collection by the corporations that develop these technologies have been sparked by the extensive adoption of FaceApp, particularly by young people on social media [251]. A family of statistical models known as "generative models" generates novel data instances that mimic training data collection. They are frequently compared to discriminative models, which can distinguish between two distinct data examples. A discriminative model might differentiate between a dog and a cat, while a generative model could create fresh images of cats that resemble real cats. The term "generative model" refers to a variety of models, including Autoregressive models like Generative Adversarial Networks (GANs), and PixelRNN, Variational Autoencoders (VAEs). Recent advancements in GANs, among other generative models, have shown notably high accuracy in synthetic media production and have been widely used in popular media. With a particular emphasis on GANs, we study the applications of several generative models in this paper [252].

Y. Liu, Zou, et al. review Generative Artificial Intelligence's (GAI) impact on materials research due to its content generation abilities. Through adopting the Prompt paradigm and reinforcement learning (RLHF), GAI transitions from specific to broader patterns, addressing complex tasks like understanding structure-activity relationships. Their

comprehensive review evaluates GAI's development, assesses various model strengths and weaknesses, and explores specific applications such as materials design and data augmentation. Using ChatGPT as an example, they showcase GAI's potential in generating diverse materials content and solving complex problems. Additionally, they identify six challenges in GAI's use in materials science and propose solutions to improve materials data generation for accelerated research and development [253].

Material characterization and data analysis

Analyzing experimental or simulated data to understand material characteristics and behaviors, optimizing materials for particular purposes, finding novel materials, and advancing materials science research are all aspects of material characterization and data analysis in machine learning [254,255]. They are essential to advancing materials engineering and make it possible to create better materials for various applications. Since short training sets are frequently used for ML models, it isn't easy to replicate the gains made in other fields. Applying ML algorithms as a rapid fitting method for short, low-dimensional datasets is possible [256]. By limiting the configurational space of materials and concentrating on specific semiconductor groups characterized by unchanging composition or crystalline structures, ML models may be built to handle the problem [141,257]. Another strategy suggested by Zhang et al. [258] involves adding rough property predictions in the feature space, improving prediction accuracy with less degree of freedom and more minor materials data.

In conclusion, combining chemical (dis)similarity principles with machine (or statistical) learning methods has demonstrated effective and accurate prediction for various properties in material systems that are not directly linked. Using machine learning, CNOHF can go through how energetic molecule attributes can be deduced from their molecular structure. Active materials have been screened using thermochemical and prohibitively expensive quantum simulation predictions [259]. Compare various molecular feature engineering methods in-depth, including custom descriptors, sum over bonds, Coulomb matrices, fingerprints and Bag of Bonds. Let's talk about the ideal technique to describe aggregate over bonds (bond count), and the perfect model was kernel ridge regression. Considering the short sample size, we can predict explosion pressure, explosive energy, explosion velocity, density, the heat of creation, and other out-of-sample parameters with acceptable errors and Pearson correlations [259]. While convergence with a high molecule count is sluggish, it's noticeable that the error can be reduced by introducing a distinct dataset, including approximately 300 extra molecules during training.

Machine learning in image processing

Utilizing machine learning for the processing of images is opening doors to discoveries day by day in medical science, material science, autonomous vehicles, security and surveillance, environmental monitoring, agriculture, etc. [260]. This ability to achieve noteworthy results across diverse domains is attributed to substantial progress in data accessibility and increased processing power among professionals. Modern machine learning systems can process image data similarly to the human brain's understanding of images. Numerous strategies are currently being used, including feature extraction, segmentation, identification, and classification, which have changed with the discovery of new dependencies when utilizing machine learning [261]. The capacity of deep learning to autonomously learn hierarchical representations from raw image data has significantly impacted image processing activities, transforming the industry. Many extremely distinct items have very different structures in natural images [262]. This enables the network to learn incredibly sophisticated and various filters, particularly in the deeper layers. One of the promising fields with a focus on medical image analysis is deep learning [263]. Different forms of image

processing are shown in Fig. 7.

Image classification

In machine learning, training a model to automatically categorize or classify an input image based on its visual information is known as image classification [264]. There are typically two steps to machine-learning-based image classification models: the training and classification phases. The system utilizes a Machine-Learning Algorithm for updating classifier model parameters during the training phase [265, 266].

Fig. 8 denotes both the training and classifying phases employ the feature extraction component. The image classifier receives its inputs from the feature vectors of the images. After that, it gives the actual results [265].

Object detection and localization

Finding an object in an image and localizing it with a bounding box is a procedure known as detection or localization [267]. Single-class object detection involves recognizing one specific object instance in an image, whereas multi-class object detection involves identifying the classes of all objects present [268].

A technique of Object Detection and localisation has been applied to articulated objects like people [269,270] and stiff objects with a lot of movement, such as faces and cars [271–274]. Object identification requires handling various problems, including partial/full occlusion, shifting lighting conditions, postures, scale, etc. Object detection is the first step in any visual recognition activity, as seen in Fig. 9 [268].

The field of computer vision involves a variety of object detection and localization algorithms and methods [268]. Fig. 10 shows object detection and localization techniques.

Image segmentation and object recognition

The process of segmenting an image separates it into various portions. It divides the image into segments made up of pixels, with each segment serving as a boundary or mask that designates its place or function within the image [275,276]. This mask divides the image into segments represented by the quantity and variety of segments. Generally speaking, supervised and unsupervised segmentation approaches fall into two groups but due to their quick and dependable performance, semi-supervised segmentation approaches have attracted a lot of interest [277]. In contrast, supervised segmentation algorithms use training examples to include prior knowledge about the image-processing task [278]. The 1990s saw a lot of interest in supervised models, such as atlas-based segmentation techniques [279,280]. Several IT disciplines utilise the image segmentation method, including computer vision, robotics, medical imaging, security procedures, and many more.

Object recognition involves identifying objects or items within an image [281]. It uses the inherent characteristics of detected objects in the image to establish familiarity. Machine learning models are employed to recognize objects within images and represent them in a manner that allows each object to comprehend its position or role [282]. It is used in image processing, video analysis, robotics, security processes, user interface design, and many more.

Image enhancement and restoration

Machine learning approaches for improving image quality, denoising, and repairing damaged or blurry photographs include image enhancement and restoration [283,284]. Deep Reinforcement Learning has found application in various tasks related to image processing. [285–287]. However, these techniques are restricted to simple applications, such as image cropping [284] and global colour improvement [288], and can only carry out global changes for the entire image.

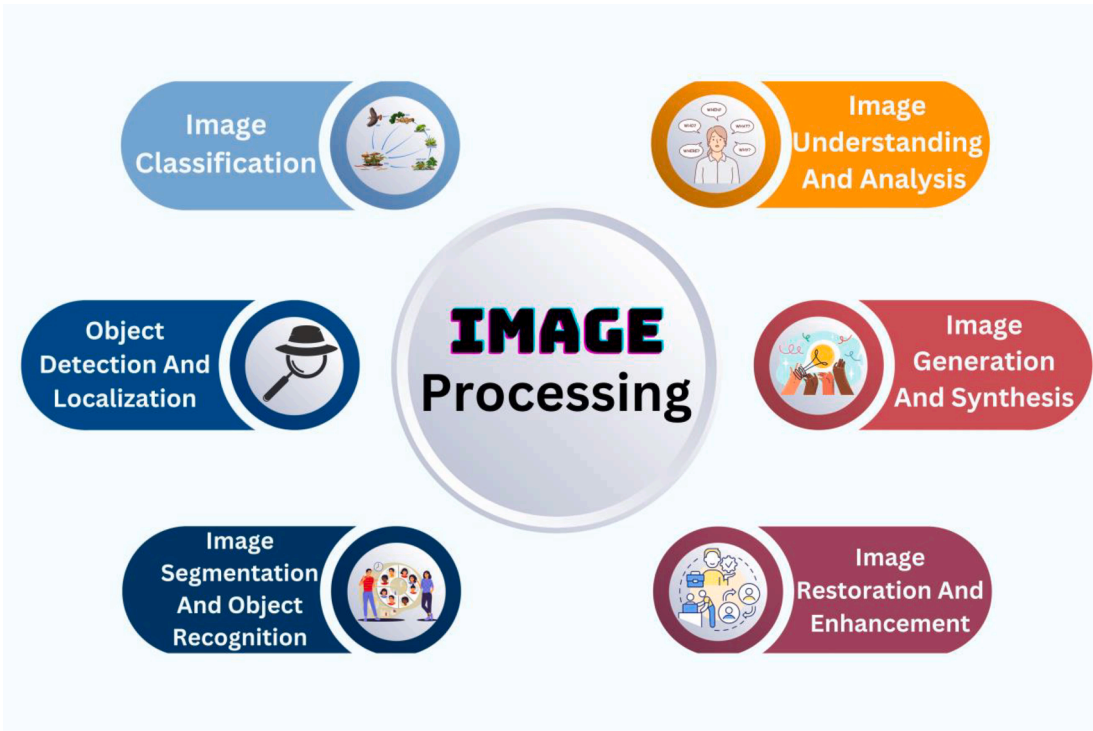


Fig. 7. Different types of image processing.

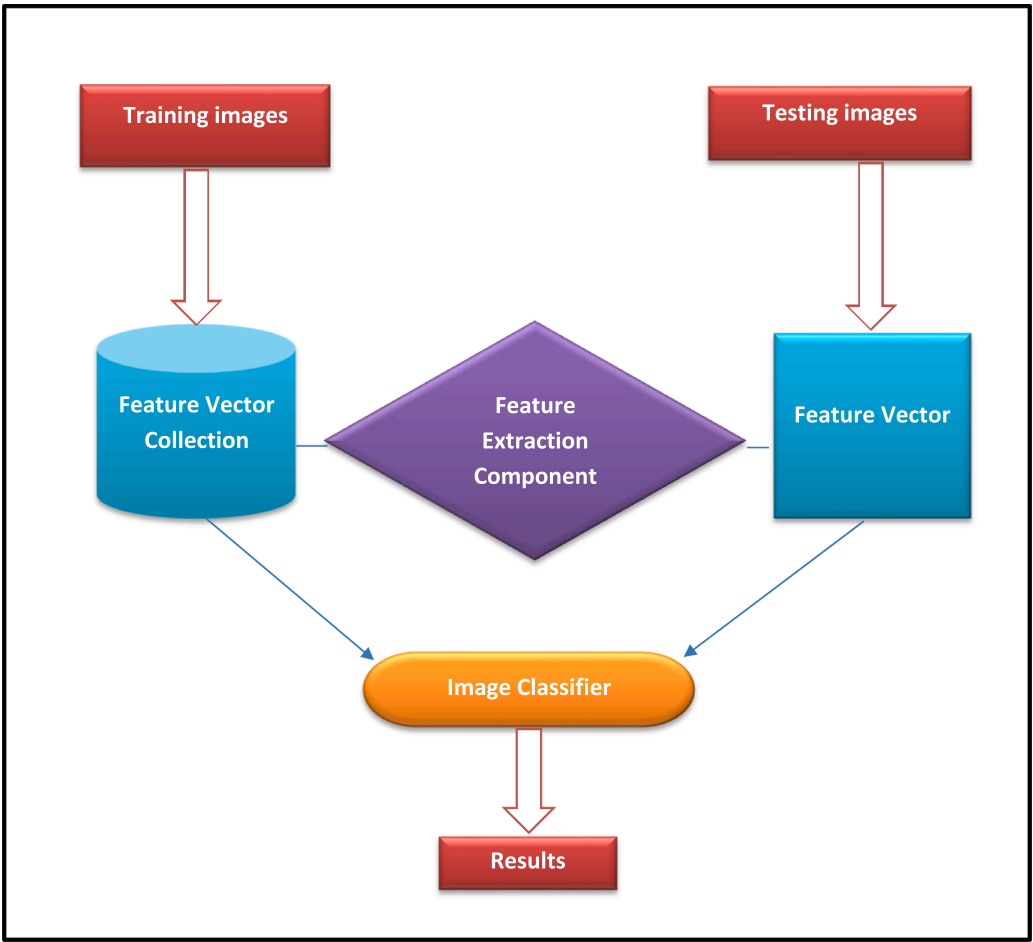


Fig. 8. Image classification architecture.

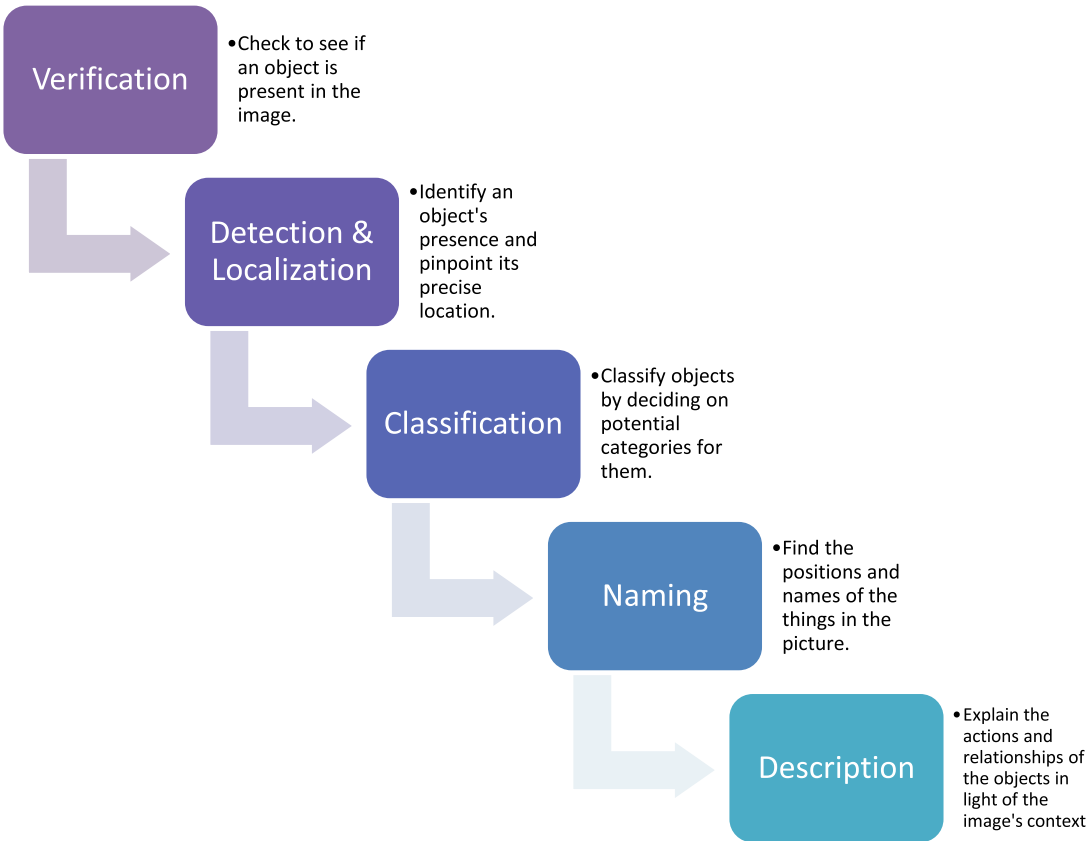


Fig. 9. The initial stage of visual recognition is object detection.

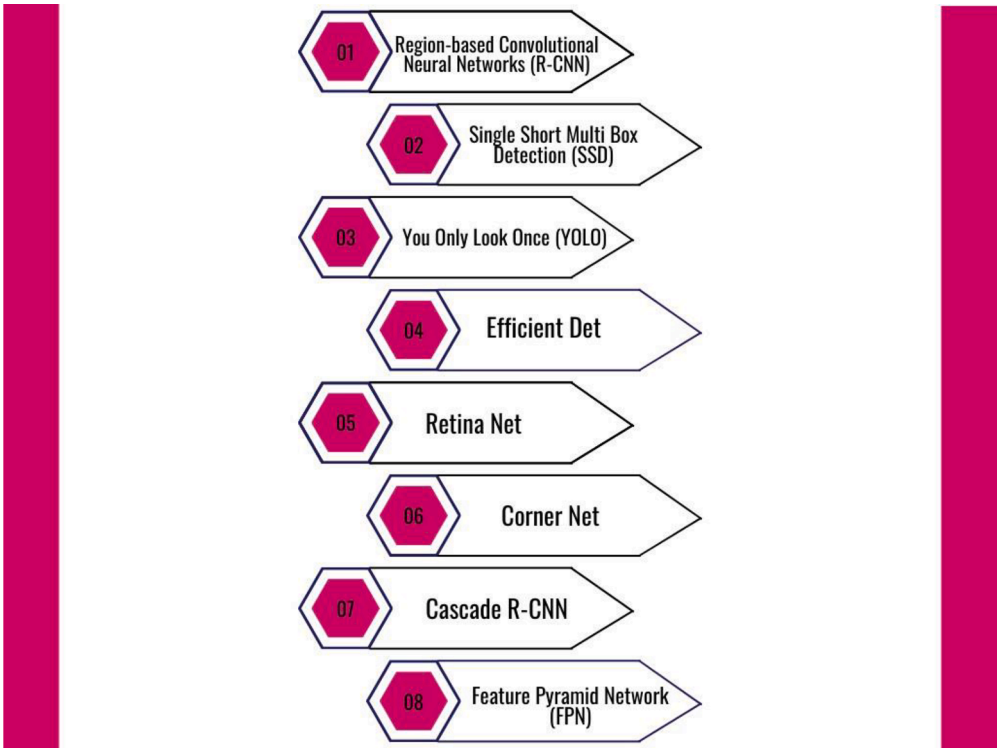


Fig. 10. Object detection and localization techniques.

However, there are still few deep reinforcement learning (RL) applications for image processing [283]. Enhancing an image to improve its qualities or attributes is known as image enhancement. Improving noisy images requires the real-time design of hardware and algorithms [289]. The development of several real-time enhancement algorithms has included techniques like CLAHE, Retinex [290,291], colour correction [292–295], and classification models [296–299]. This procedure is carried out automatically to enhance image quality using deep learning.

Image restoration involves repairing a damaged image, aiming to restore its original quality while safeguarding the unaffected parts of the image. The concepts, images, or pieces lost in a damaged image can be recovered using machine learning [300]. It evaluates the optimal qualities in the prior state for damaged images and displays their advantageous characteristics from that condition [301]. Several methods, including CNNs, autoencoders, and Generative Adversarial Networks (GANs), are used for image restoration [302]. In underwater image restoration, the atmospheric deterioration model developed by Chiang and Chen (2012) is frequently utilized [303]. The CNN technique enhances the image's contrast and colour cast [301]. This method is used for recovery, image restoration, character change correction, etc.

Image generation and synthesis

Image synthesis for machine learning effectively creates large training data sets, which guarantees optimal distribution, content variety, and the potential to be a crucial component in deep learning training pipelines [304]. Enhancing image generation in the context of machine learning, it is necessary to combine and classify image synthesis methods in the context of image generation, using a taxonomy based on modelling and rendering, and to categorize their application in computer vision applications with a particular focus on computer graphics [304]. The potential for deep learning approaches, particularly convolutional neural networks (CNNs) and generative adversarial networks (GANs), Cycle-Consistent Adversarial Networks (CycleGAN), U-Net, and AE (Auto-encoder) to combine training data is enormous [305–308]. Label-to-MRI translation (synthetic image synthesis) and MRI-to-label translation (image segmentation) are both accomplished using the image-to-image translation conditional GAN (pix2pix) model proposed in [309,310]. AE (Auto-encoder) uses convolution kernels to investigate

spatially local visual patterns [274]. GANs are composed of a generator and discriminator neural network, where the generator creates synthetic imagery from random vectors, and the discriminator tells the difference between them. This dynamic forces the generator to refine synthesis and produce more realistic imagery [308]. Halder et al. [311] presented a face synthesis method based on geometry modelling, and Patel and Smith [312] used the information of the obscure section as an auxiliary condition to generate a three-dimensional morphable model, both of which were inspired by widely used and achievements in computer vision leveraging geometric knowledge. A technique by Ramalingam and Viet [313] focused on using affordable 3D sensors to synthesize faces to construct or increase the face recognition database. Several works that include statistical learning theory have also been proposed simultaneously [314]. Enhancing their future performance can be accomplished by integrating more data, such as physical restrictions, demographic data, and particular scene information, which has been especially evident in the rapid dominance of [304–307].

Image understanding and analysis

Utilizing machine learning methods to identify features and their corresponding data from images for interpretation and analysis. It is capable of locating image components and analyzing or evaluating image attributes. Convolutional Neural Networks (CNN), Residual Neural Networks (ResNet), Advanced Convolutional Neural Networks for Machine Vision, etc., are examples of technologies that can be utilized for image understanding and analysis [315]. Most machine vision and image understanding systems have two main steps, as depicted in Fig. 11. The initial step involves taking the picture data and extracting an appropriate and effective form of knowledge representation. The real-time examination and processing of the image are done in the second stage, typically on the extracted representation [315].

Recent advances of machine learning in materials research

Rapid advancements in energy, information science, national defence, and other areas have placed significant and varied demands on materials. However, conventional methods for finding novel materials, such as the empirical trial and error method and the density functional

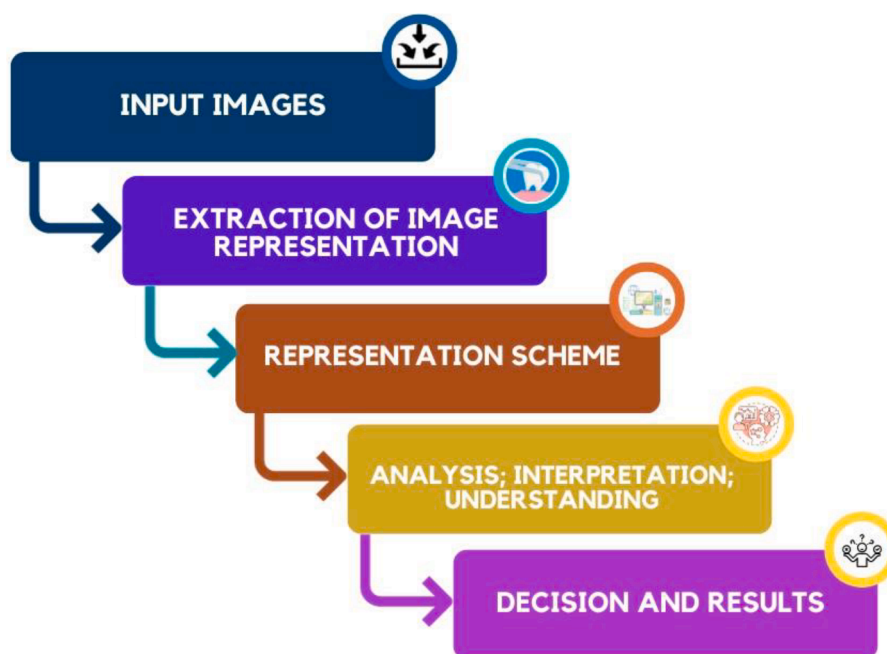


Fig. 11. An image understanding and computer vision system's fundamental components.

theory (DFT)-based method, typically need a lengthy R&D cycle, are expensive and inefficient, and have trouble keeping up with the current state of materials science. Machine learning is one of the most effective approaches to replace repeated laboratory tests or DFT calculations since it can significantly lower computational costs and expedite the development cycle [316]. In recent materials research, machine learning uses historical data to forecast phase diagrams, crystal structures, and properties with faster and more precise predictions. Also, developing interatomic potentials, real-time analysis of experiments, and mapping complex behavior to process variables [317]. Material scientists increasingly employ machine learning to detect patterns and predict outcomes from data [318]. The field of materials science has a rising opportunity to use and expand upon the developments in artificial intelligence (AI) and machine learning (ML) that are reshaping and restructuring sectors throughout the economy. There is reason to believe that these developments will speed up the discovery, creation, and application of novel materials, bringing materials Scientists one step closer to attaining the goals of the Materials Genome Initiative [318]. Over the past ten years, a rapidly expanding discipline has used ML to design and develop problems in materials science and other fields [319]. ML is unquestionably altering the entire paradigm in practically every sector, including science and engineering. Recent spikes in interest in machine learning (ML) may have led some people to believe ML is a universal fix for all problems.

ML may be a helpful strategy for an issue that humans frequently struggle with, such as when the data and their interactions are too complicated and cannot be understood. In contrast, a human could thrive with sparse data, such as determining from a small set of given data that ML cannot accomplish well. Therefore, examining whether the question can be answered using ML before using it in research is essential. The three steps of the ML method are typically as follows [320]. Fig. 12 shows the elementary stages of applying machine learning in materials science.

Microscopic property prediction

A material's microscopic features, such as its atomic and structural traits, such as the lattice constant, influence its macroscopic performance. As far as we know, machine learning techniques for predicting microscopic properties focus on several factors, such as band energy, lattice constant, molecule atomization energy, and electron affinity. One of the primary hurdles in developing new substrate or buffer materials for semiconductor epitaxy is the lattice mismatch between layers of different materials. The prediction of lattice constants for various element combinations can be expedited using lattice constant prediction. This approach is becoming essential for advancing high technology, as it enables rapid and accurate estimation of lattice constants for numerous unidentified compounds. Machine learning, particularly regression analysis, offers a viable solution for this prediction challenge [321].

Material characterization and imaging

Advances in high-resolution imaging techniques have created new opportunities for machine learning (ML) approaches to swiftly and quantitatively analyze functional materials statically and dynamically. This complements traditional methods like modelling, simulations, and synthesis. Historically, characterization focused on qualitative aspects, but contemporary imaging tools, with enhanced resolution and multi-probe capabilities, provide highly detailed quantitative information. Modern real-space imaging methods, such as scanning transmission electron microscopy, scanning tunneling microscopy, and atomic force microscopy, enable direct visualization of atomic-level structures and functional features in complex materials with multiple components and phases [322–326].

Recent advancements in materials imaging and characterization methods enable the acquisition of high-quality data for functional materials under static and dynamic conditions. These developments stem from cutting-edge techniques such as scanning transmission electron microscopy, tunnelling microscopy, atomic force microscopy, and

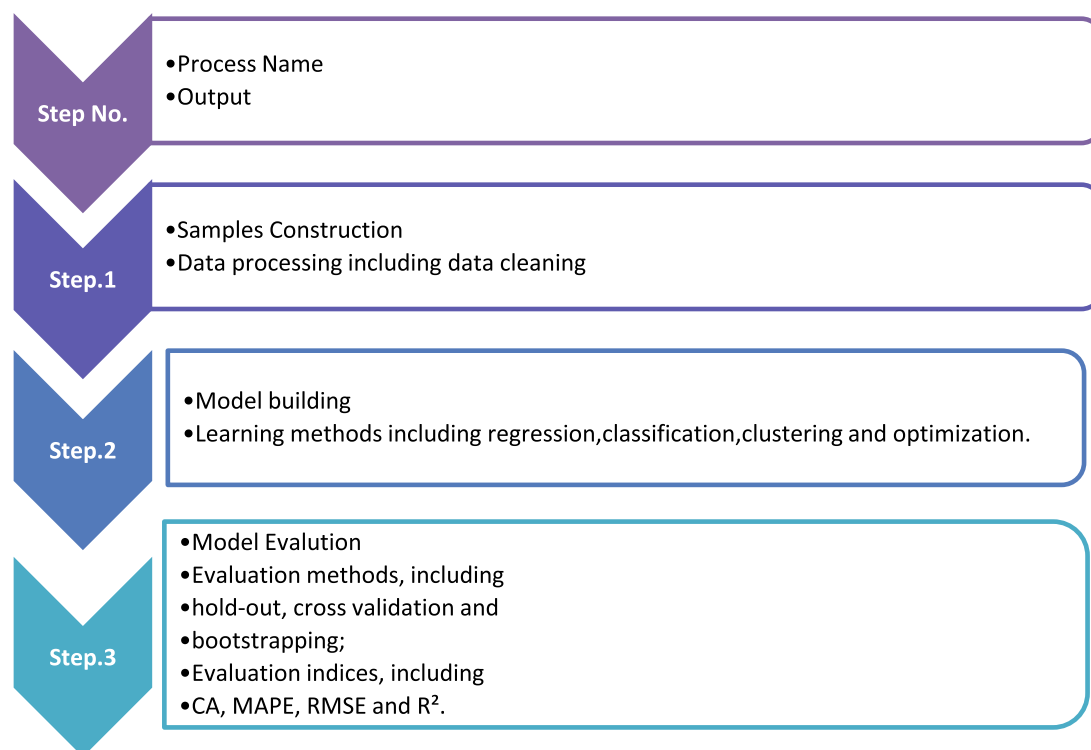


Fig. 12. The general process of machine learning in materials science.

electron backscatter diffraction imaging. Concurrently, enhancing machine learning-driven image recognition offers real-time characterisation data analysis opportunities, reducing workflow bottlenecks in materials development and facilitating knowledge extraction from the data using physics-informed models (Ma et al.) [327]. An ML approach was used to map microstructure images to processing conditions in the context of uranium-molybdenum nuclear fuel alloy. The method achieved strong classification accuracy, with an F1 score of 95.1 %. Using a novel microstructure representation, it successfully differentiated between micrographs related to 10 distinct thermo-mechanical material processing conditions [328]. Ziatdinov et al. used a Gaussian process regression-based extrapolation strategy for the contact Kelvin probe force microscopy method over the spatial and parameter space. The authors also include an interactive Google Colab notebook that describes every aspect of the analyses included in this study as their technique implementation. ML-assisted reliable data is a challenge that Vasudevan et al. [329] experienced. Through a concrete illustration of Bayesian inference applied to studying image generation mechanisms in a ferroelectric thin film, we enhanced the trustworthiness of interpreting intricate scanning probe microscopy data. The utilization of the Bayesian framework highlights its ability to integrate prior material knowledge and facilitate model selection.

Material property analysis

Machine learning methods are applied to predict and understand material properties by analysing their characteristics [330]. Machine learning algorithms can find patterns, correlations, and predictive models using large datasets of material compositions, structures, and associated property measurements. These models can speed up identifying and improving materials for particular applications, such as finding new catalysts, boosting mechanical strength, or increasing thermal conductivity [331]. Overall, machine learning makes it possible to design materials more effectively and intelligently, which speeds up innovation across numerous industries [332].

Liu et al. [333] underscore machine learning's pivotal role in accelerating the discovery and design of electrochemical energy storage materials, especially in rechargeable batteries. The review highlights ML's state-of-the-art applications, emphasizing its use in feature selection methods to extract crucial material properties. It explores ML's diverse applications in predicting properties and discovering new battery materials, while also pinpointing challenges such as high dimensionality versus small sample sizes, model complexity versus accuracy trade-offs, and potential discrepancies with domain expertise. It proposes strategies to tackle these hurdles and outlines future research directions, showcasing ML's immense potential in optimizing rechargeable battery materials. Liu et al. [321] review how machine learning revolutionizes materials science by significantly expediting the discovery and design of new materials. Its application streamlines processes, enhancing prediction accuracy and time efficiency. The review outlines the basic procedures of ML, algorithmic comparisons, and its diverse applications in predicting material properties and facilitating discoveries. Despite its advantages, challenges such as interpretability and data quality persist. Future efforts aim to integrate computational and experimental approaches for more targeted and efficient material research, ultimately enhancing our understanding of material properties.

Shi et al. were the first to identify three primary contradictions that hinder the effective application of ML within the material community: conflicts between learning outcomes and domain knowledge, between model complexity and user-friendliness, and between dealing with high-dimensional data and limited sample sizes [334]. This work effectively reconciles the aforementioned contradictions by incorporating domain knowledge from material science into ML methods. It specifically utilizes crystal structure, bond length, and the coordination environment of local atoms as structural descriptors to establish mapping relationships

between structure and energy (force). The study explores the fitting of MTPs across various levmax values and identifies their performance in predicting lattice constants, volumes, and mean absolute errors (MAEs) in energies and forces [335]. To tackle these, the study integrates material science expertise into ML methods, employing crystal structure, bond length, and atom coordination as descriptors. It explores various parameters' impact on predictive performance, like lattice constants, volumes, and error metrics. Addressing these challenges is crucial for advancing ML's effectiveness in materials research.

Liu et al.'s study introduces a novel method called DML-FSdek for selecting material features in machine learning models, addressing challenges in feature selection. Unlike traditional methods, this approach automates the process, integrating domain expert knowledge to avoid crucial feature omission. It demonstrated success in ten material properties datasets, automatically identifying a smaller yet more relevant feature set. This streamlined approach establishes accurate structure-activity relationships, ensuring interpretable results without sacrificing prediction accuracy. In essence, DML-FSdek achieves a more focused material property representation through Multi-Layer Feature Selection [334]. Fig. 13 illustrates different applications of material property analysis.

Degradation detection

Machine learning surpasses human judgment in precision and practicality when analyzing materials for purposes like identifying metal corrosion, detecting cracks in asphalt pavement, and assessing the strength of concrete [336–340]. In their work from 1996, Agrawal et al. examined how the properties of an alloy, along with its composition and manufacturing process parameters, are connected. They explored the application of machine learning techniques, including feature selection and predictive modelling, to forecast the fatigue strength of steel. The tempering temperature was shown to be the most significant factor influencing fatigue strength out of the 25 features examined. The process has four steps: The raw data were initially preprocessed with domain expertise. Second, high-relevance features were chosen using ranking-based feature selection techniques. The fatigue strength was then predicted using machine learning algorithms. Finally, Leave-One-Out Cross-Validation (LOOCV) was employed to assess the models' accuracy [341].

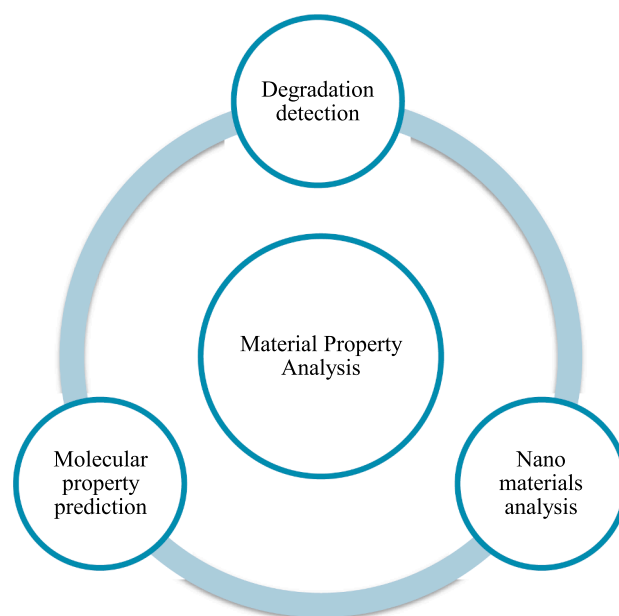


Fig. 13. The application of the material property analysis.

Nanomaterials analysis

The importance of machine learning is increasing significantly within the realm of nanomaterials as artificial intelligence advances [342–345]. Applying machine learning to research C60's solubility dates back to 1993 [346]. The prediction of nanomaterial toxicity, the discovery of novel nontoxic nanoparticles, the development of multi-structure/single-property relationships of NPs, the study of quantum-mechanical observables of molecular systems, the analysis of chemical reactions of nanomaterials, and the solution of kinetic systems have all benefited from the widespread use of machine learning [342–345,347–352].

Molecular property prediction

Utilizing high-throughput density functional calculations to forecast molecular properties take a lot of time. Machine learning is a practical alternative that can achieve high accuracy for quick prediction of the structures or properties of materials, compounds and molecules [353–357]. ElemNet is a DNN-based model that uses elements as input to forecast the properties of materials. It swiftly produces precise predictions by automatically extracting the physical and chemical relationships and similarities between elements. Additionally, Ward et al. suggested a broad machine-learning paradigm for forecasting inorganic material properties. This model can predict various characteristics of both crystalline and amorphous materials, such as band gap energy and the potential for forming glass [358].

Material discovery and designing

Machine learning (ML) has extended its capabilities in predicting properties, enabling the discovery, design, and production of innovative materials across various applications and categories. It provides new insights into the underlying chemical or physical relationships that govern desired attributes. For example, in the domain of halide perovskites for solar photovoltaics, ML has revealed trends in properties (e.g., bandgap and stability) based on factors such as halogen content and alkali or organic species. This has led to the identification of promising new materials like Cesium Aurous Iodide ($\text{Cs}_2\text{Au}_2\text{I}_6$) and Ammonium Amide Indium III bromide ($\text{NH}_3\text{NH}_2\text{InBr}_3$), with the former being extensively studied as a potential solar material [359–362]. Furthermore, employing machine learning to predict materials data across a wide array of elpasolite compounds based on Br and Cl led to the discovery of several promising novel scintillator materials. Over 20 established scintillators were also successfully replicated. Notably, the favourable positioning of Cerium III (Ce^{3+}) 4f and 5d energy levels

within the material bandgap was achieved, fulfilling a crucial design requirement for scintillators. These compositional improvements were enabled by the valuable insights gained from machine learning [363]. Fig. 14 illustrates the schema of high-throughput experiments for contemporary materials.

Challenges and prospects

Challenges

Materials science has evolved beyond three approaches, encompassing mathematical theory, simulations, and experimental research [364]. The first-paradigm investigation has no scientific quantitative basis and entirely relies on the observer's intuition. The development of physical models denoted by mathematical calculations, which gave rise to some theoretical underpinnings like the thermodynamics principles for materials study, began only a few centuries ago [365]. As a result of the development of computers decades ago, which enabled the simulation of challenging real-world issues using the knowledge gathered from the second paradigm, the third scientific paradigm is now emerging. Two notable approaches to materials research using big data computing—density functional theory and molecular dynamics—have been proposed [366].

Y. Liu, Ge, et al. developed an automatic descriptors recognizer using NLP for mining descriptors from materials science literature. Their system, integrating a conditional data augmentation model (cDA-DK) and coarse- and fine-grained descriptor subrecognizers (CGDR and FGDR), significantly reduces manual labeling costs. CGDR achieved an 87 % accuracy in recognizing coarse-grained descriptors, while FGDR screened 408 descriptors. They constructed six models with a 96 % prediction performance for activation energy, showcasing the potential of automated descriptor extraction to advance materials design and discovery by understanding structure-activity relationships [367].

Nevertheless, the transition from research to practical application in traditional materials science often spans 15–25 years or more, with some promising findings never actually being implemented due to perceived inaccuracies [368]. The prevalence of trial-and-error methods is a significant factor. As materials research deals with extensive data and intricate dimensions, material characterisation techniques are growing [369,370].

Y. Liu et al. present NCOR-FS, a feature selection method using materials domain knowledge to improve feature quality for ML models predicting materials properties. It incorporates Non-Co-Occurrence Rules (NCORs) derived from correlated features, guiding subset

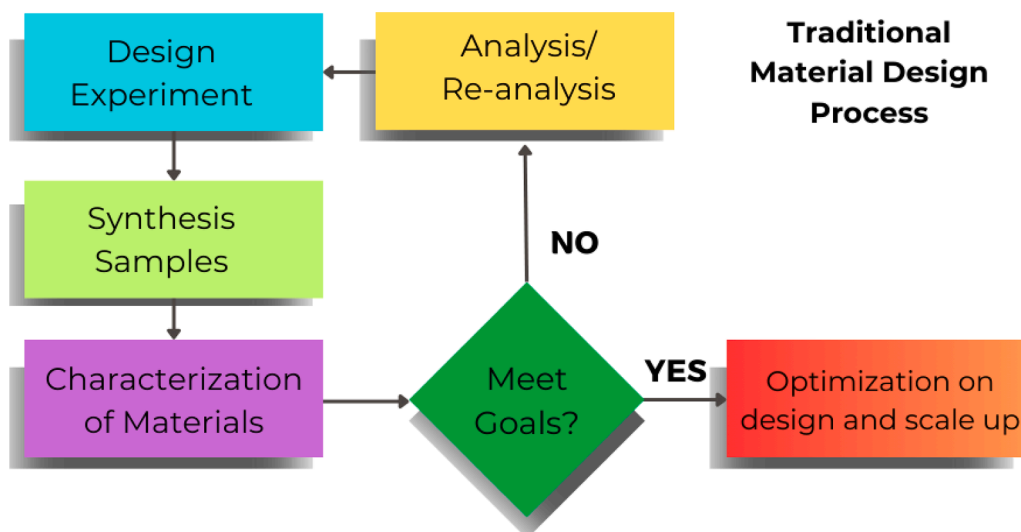


Fig. 14. Traditional Material Design Process.

selection via a swarm intelligence algorithm. Compared to other methods, NCOR-FS enhances prediction accuracy and model interpretability across seven materials science datasets, showcasing the value of domain knowledge in constructing robust ML models [253].

Both macroscopic and microscopic descriptions can be used to characterize a material's characteristics, such as melting point, hardness, glass transition temperature [371,372], ionic conductivity, lattice constant, and molecular atomization energy. Computational modelling and experimental measurement are the most popular techniques for examining material properties. These two techniques entail intricate procedures and experimental setups. As a result, it is challenging to create computational simulations that accurately represent the complex logical connections between a material's qualities and the causes that affect them, some of which may even be unknowable [373]. Additionally, investigations conducted to gauge a compound's characteristics typically occur toward the end of the materials selection process. As a result, the significant time and extensive experimental resources invested will be wasted if the outcomes are unsatisfactory. Investments made up to that time are ineffective. Furthermore, despite enormous computational or experimental efforts, exploring the properties of materials often poses significant challenges or maybe practically unfeasible. The rapid development of intelligent and high-performance prediction models is crucial for accurately predicting material properties efficiently, both in terms of time and computational resources [373].

Prospects

Although calculations and predictions of material structures and performance can be achieved at various scales using first principles, phase-field theory, molecular dynamics, and finite element analysis (FEA), these models are usually tailored to particular material systems. It can still not satisfy the demands of several descriptions for various qualities, which severely limits future material research. These issues are starting to be solved as artificial intelligence is promoted and made more widely known, bringing materials research to the fourth scientific paradigm. The fourth paradigm of materials science, also known as big data-driven science, is currently emerging in materials research, which has profited from the massive amounts of data produced by experiments and computation techniques like first principles and molecular dynamics theory [374]. Additionally, increased applications of high-throughput computers in materials research are causing the data to grow quicker than ever [368]. The emergence of the artificial intelligence (AI) revolution has led to a growing focus on using machine learning (ML) for predicting material properties and discovering novel materials, making it a vibrant area of study [353].

Machine learning has found widespread application in predicting properties, exploring novel materials, and delving into quantum chemistry due to its robust predictive capabilities and cost efficiency. However, employing machine learning in materials science poses several challenges. For instance, the scarcity of high-quality materials data hinders accurate property descriptions, and prediction accuracy lags behind that of Density Functional Theory (DFT) calculations. This article proposes additional avenues to enhance the utilization of machine learning in the materials science domain.

First, to foster the future development of machine learning, it's essential to prioritize rapidly creating a comprehensive material database. Machine learning heavily relies on data, and the quality and quantity of data directly influence its accuracy. Valuable material data, encompassing molecular properties, reaction conditions, and synthetic compositions, are scattered across scientific literature and experimental records. Leveraging text mining can efficiently gather this valuable information from papers, journals, and magazines, significantly expanding existing material databases and creating specialized ones [316].

Second, it is essential to innovate novel machine-learning principles. As time progresses, enhancing deep learning and substituting manual feature engineering will lead to more efficient representations of raw

data. However, researchers are still unsure how DNNs choose their features and what those features represent. As a result, the deep learning results are not sufficiently convincing, and no broadly applicable theory is produced. In addition to improving the applicability of machine learning in materials science, trying to comprehend what is happening within the "black box" makes it easier to find natural laws that are unknowable to humans [316].

Third, a significant application of machine learning may be in quantum chemistry. Numerous issues in quantum chemistry can be resolved using machine learning thanks to its robust data-processing capabilities. The model's prediction accuracy can be significantly improved by integrating DFT and machine learning. This could be an effective technique for forecasting complex molecular features and architectures, researching quantum multibody systems, and finding novel [355].

Limited material data remains a challenge for machine learning in achieving the desired accuracy for certain tasks. Thus, there's a pressing need for highly accurate models trained on small yet reliable datasets. A deep learning model's adequacy has been demonstrated with as few as 4000 samples [375]. Furthermore, training models using failure data from unsuccessful experiments, as previously described, can be beneficial in such scenarios [376].

One promising approach is high-throughput computational materials design, a burgeoning field in materials science. Researchers employ advanced thermodynamic and electronic-structure techniques, intelligent data mining, and database construction to explore vast data repositories to discover new materials. This review provides a snapshot of this rapidly evolving area, emphasizing the forthcoming challenges and potential opportunities [377]. Computational methods are increasingly vital in materials science research. They play a pivotal role in identifying and enhancing materials for specific purposes, complementing experimental techniques. The *in silico* materials design field has thrived in the past decade due to exponential increases in processing power, algorithmic progress, and the availability of open-source scientific software [378–380]. More recently, the growing use of artificial intelligence and machine learning-based techniques has created new opportunities for designing and developing materials [381,382].

This Special Issue seeks to gather the latest advancements in computational materials design, showcasing diverse techniques from fundamental computations to molecular dynamics, mesoscale methods, and machine learning-based models for predicting materials properties. The contributions cover a broad spectrum of approaches in this dynamic field [383].

Conclusion

Incorporating machine learning into materials research has ushered in a transformative era, fundamentally reshaping how we approach and advance our understanding of materials. Machine learning is a game-changing tool that can quickly analyze complex material properties, predict novel materials, and optimize fabrication processes. Thanks to various machine learning techniques, including unsupervised, supervised, reinforcement learning, and deep learning, computers can learn from data and make wise decisions. This toolkit encourages innovation within industries and pushes the boundaries of materials science.

Machine learning significantly impacts various domains, revolutionizing material synthesis through predictive modelling, structure-property correlations, and data-driven exploration. Additionally, it enables precise predictions of material behavior, facilitating customized design and speeding up discovery. The combination of machine learning and image processing improves object detection, classification, and segmentation, revolutionizing the processing of visual information in a variety of industries. Recent developments enable accurate property prediction and data extraction at the atomic level, opening the door for previously unheard-of material insights.

Despite its potential, issues with data quality, algorithm complexity,

and ethical considerations are acknowledged. Machine learning will be essential in shaping the future of materials research because of its potential to reveal novel insights, speed up material innovation, and overcome obstacles. This thorough analysis highlights the importance of ongoing research and development while highlighting the exciting potential of interdisciplinary collaboration and algorithmic advancements to drive the efficient integration of machine learning and, in turn, chart a dynamic course for the evolution of materials research.

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CRediT authorship contribution statement

Md Hosne Mobarak: Data curation, Investigation, Methodology, Validation, Resources, Supervision, Visualization, Writing – review & editing. **Mariam Akter Mimona:** Investigation, Resources, Visualization, Writing – original draft. **Md. Aminul Islam:** Visualization, Writing – review & editing. **Nayem Hossain:** Conceptualization, Writing – review & editing. **Fatema Tuz Zohura:** Visualization, Writing – original draft. **Ibnul Imtiaz:** Visualization, Writing – original draft. **Md Israfil Hossain Rimon:** Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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